

Customer Market Analysis

Final Project Report

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1. Project Objective

Segment customers by purchasing behavior to enable targeted marketing campaigns and improve customer retention.

2. Data Collection & Preparation

A synthetic sample dataset was generated for this project (seed=42, reproducible): 50 customers and 249 transactions across multiple product categories, with dates and order amounts. This is a representative sample, not the full production-scale dataset.

3. Methodology — SQL Analysis

SQL window functions (NTILE) were used to score each customer on Recency, Frequency and Monetary value. The three quartile scores were combined into a single RFM segment label for each customer.

```
SELECT customer_id,  
       NTILE(4) OVER (ORDER BY recency_days) AS r_score,  
       NTILE(4) OVER (ORDER BY frequency) AS f_score,  
       NTILE(4) OVER (ORDER BY monetary DESC) AS m_score  
FROM customer_agg;
```

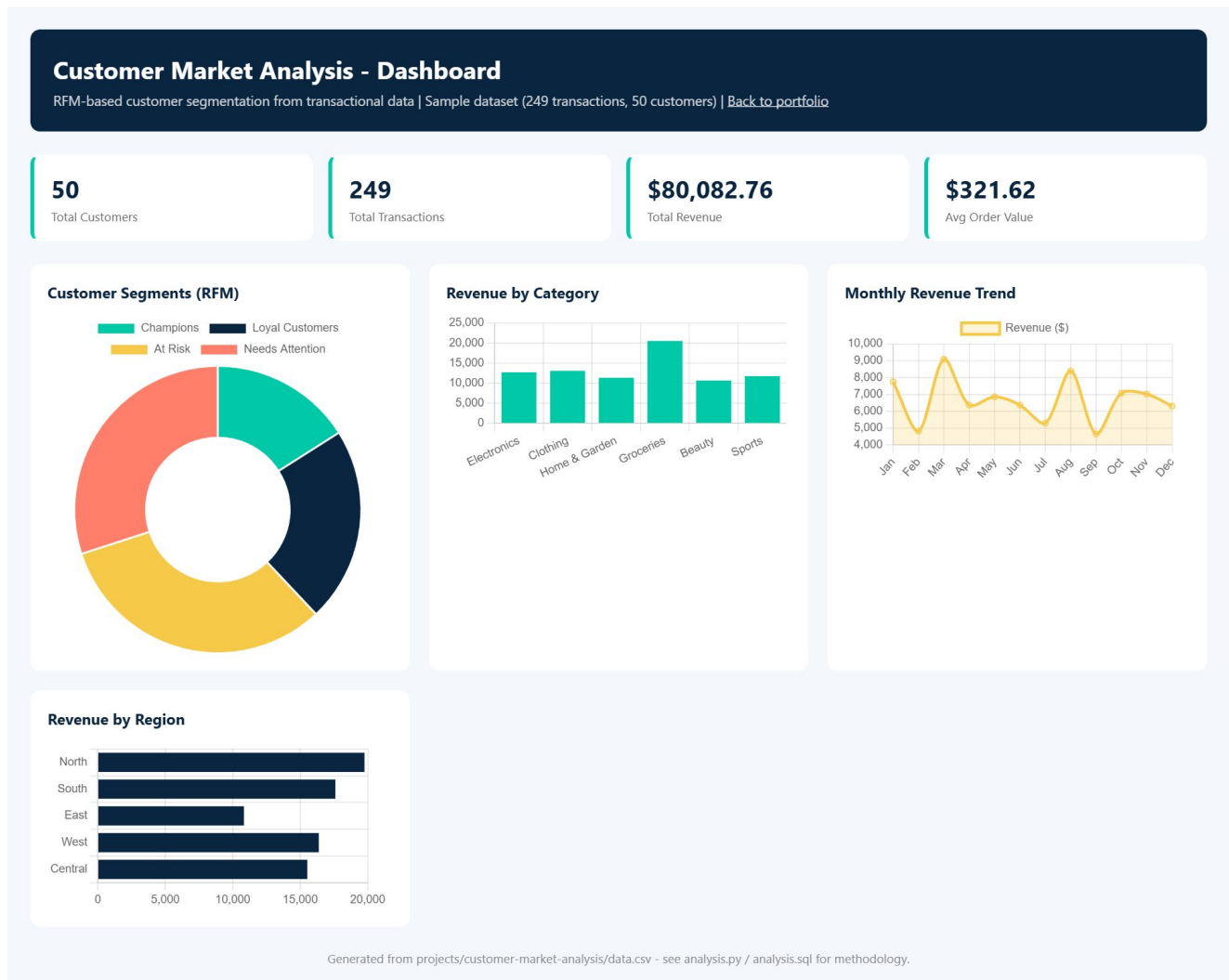
4. Methodology – Python Analysis

The RFM scores were loaded into pandas, standardized, and clustered using K-Means (scikit-learn) to group customers into behavioral segments such as Champions, Loyal, At-Risk and Lost.

```
from sklearn.cluster import KMeans  
X = scale(rfm[['recency', 'frequency', 'monetary']])  
kmeans = KMeans(n_clusters=4, random_state=42).fit(X)  
rfm['segment'] = kmeans.labels_
```

5. Dashboard Development

An interactive HTML/Chart.js dashboard was built showing KPI cards and charts (revenue by segment, top categories, RFM scatter) driven directly by the aggregated data.



6. Key Results

50

Customers

249

Transactions

\$80,082.76

Total Revenue

\$321.62

Avg Order Value

7. Insights

Champions (high value, frequent, recent buyers) drive a disproportionate share of revenue, while At-Risk customers represent a clear re-engagement opportunity for marketing campaigns.

8. Conclusion

This project demonstrates a complete customer segmentation workflow: from raw transaction data, through SQL-based RFM scoring and Python clustering, to an interactive dashboard that marketing teams can use to prioritize outreach.